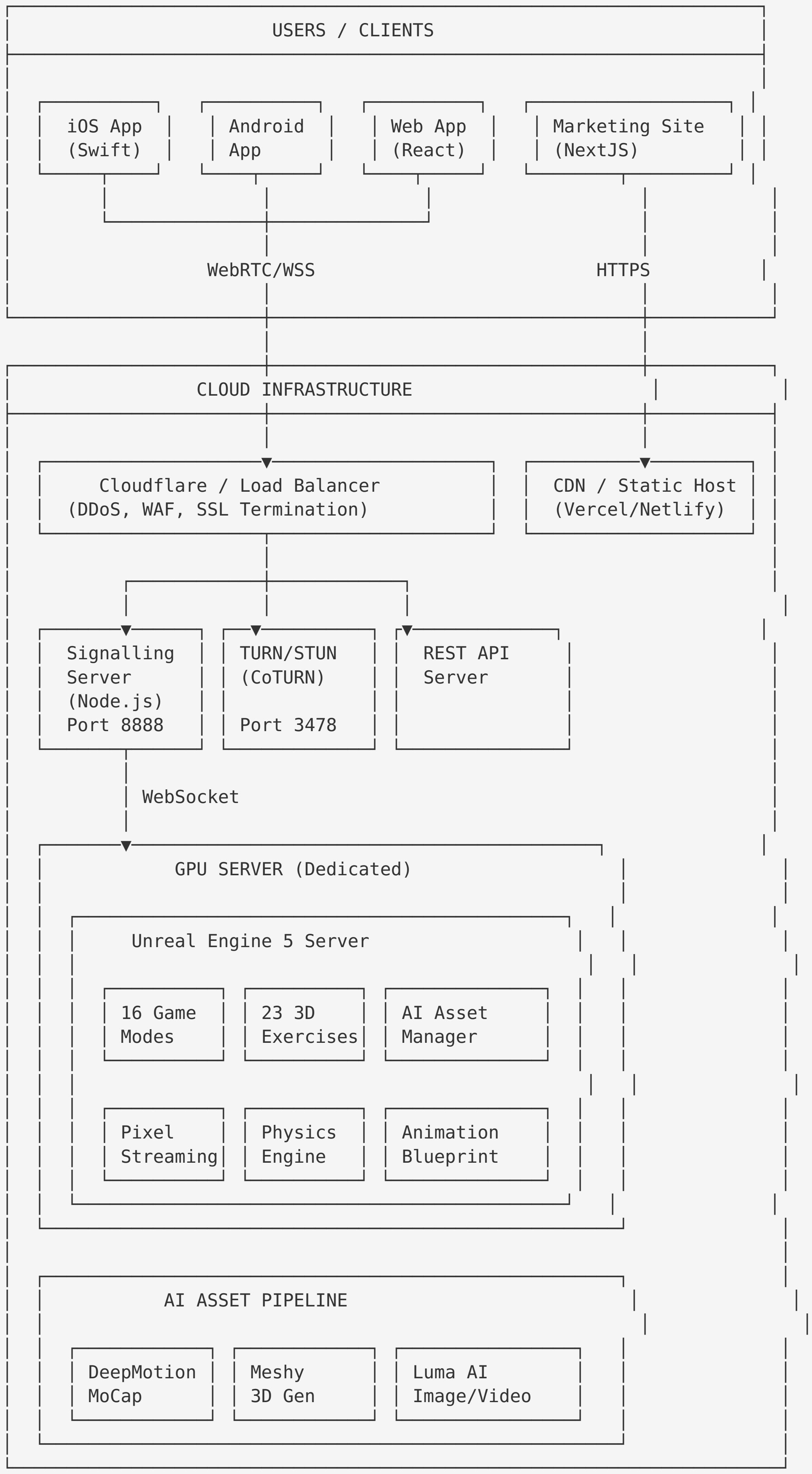
Final Evolution Lab - System Architecture

High-Level Architecture


USERS / CLIENTS
iOS App (Swift)
Android App
Web App (React)
Marketing Site (NextJS)
WebRTC/WSS
HTTPS
CLOUD INFRASTRUCTURE
Cloudflare / Load Balancer (DDoS, WAF, SSL Termination)
CDN / Static Host (Vercel/Netlify)
Signalling Server (Node.js) Port 8888
TURN/STUN (CoTURN) Port 3478
REST API Server
WebSocket
GPU SERVER (Dedicated)
Unreal Engine 5 Server
16 Game Modes
23 3D Exercises
AI Asset Manager
Pixel Streaming
Physics Engine
Animation Blueprint
AI ASSET PIPELINE
DeepMotion MoCap
Meshy 3D Gen
Luma AI Image/Video

Component Details

Client Layer

Component	Technology	Purpose
iOS App	Swift, WebRTC	Native Pixel Streaming client with HealthKit
Android App	Kotlin, WebRTC	Native Pixel Streaming client
Web App	React, Vite, TypeScript	Browser-based game client
Marketing Site	Next.js 16, Tailwind CSS	Finalevolutiongroup.com

Streaming Infrastructure

Component	Technology	Port	Purpose
Signalling Server	Node.js, Express, WebSocket	8888	Session management, WebRTC signalling
TURN/STUN Server	CoTURN	3478, 49152-65535	NAT traversal, media relay
Frontend Server	Vite/Serve	3000	Web app hosting (dev)

Game Engine

Component	Technology	Purpose
UE5 Server	Unreal Engine 5.4	Game rendering, physics, Pixel Streaming
16 Game Modes	UE5 Blueprints + C++	Basketball, Karate, Soccer, etc.
23 Exercises	DeepMotion animations	Motion-captured fitness exercises
AI Asset Manager	C++ Subsystem	Runtime loading of AI-generated assets

AI Pipeline

Service	Purpose	Assets Generated
DeepMotion	Motion capture from video	23 exercise animations, 54 athlete moves
Meshy	Text/Image to 3D	9 props, 8 environments
Luma AI	Reference images/videos	12 venue references

Data Flow: Game Session

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1. User opens app → connects to wss://stream.finalevolutiongroup.com
2. Signalling server assigns player ID → routes to available UE5 instance
3. UE5 server starts game mode → begins Pixel Streaming encode
4. Video frames: UE5 → H.264/VP8 encode → WebRTC → TURN relay → Client
5. Input: Client touch → WebRTC data channel → UE5 input handler
6. Latency budget: <30ms end-to-end
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Service URLs

Service	URL
Marketing Website	https://finalevolutiongroup.com
Web App	https://app.finalevolutiongroup.com
Streaming Signalling	wss://stream.finalevolutiongroup.com
REST API	https://stream.finalevolutiongroup.com/api/*
TURN Server	turn:turn.finalevolutiongroup.com:3478
STUN Server	stun:turn.finalevolutiongroup.com:3478